

Week 3 Customer Segmentation

Project: Customer/Data Science Internship

Path: C:\workspace\Decodelabs-Data-Science-Internship\Week_3_Customer_Segmentation

Week 3 — Customer Segmentation Report

1. Executive Summary

This project implements a customer segmentation pipeline using PCA and K-Means clustering. It includes data cleaning, feature scaling, dimensionality reduction, cluster validation (elbow & silhouette), final clustering, and business-facing customer personas.

2. Dataset Overview

- The pipeline automatically loads a dataset from `Week\_3\_Customer\_Segmentation/data/`. - Supported formats: CSV and Excel (.xlsx). - The notebook prints dataset shape, columns, data types, missing value summary, and duplicates.

3. Data Cleaning

- Missing values are handled with simple imputation (dropping or filling where appropriate). - Duplicate rows are removed. - Cleaned dataset saved to `outputs/cleaned\_customers.csv`.

4. Feature Selection

- Numeric features are automatically selected for clustering. - Identifiers (customer IDs) are removed from the feature matrix.

5. PCA Analysis

- PCA is applied to the scaled numeric features. - The explained variance plot is saved to `charts/pca\_variance\_explained.png`. - Data reduced to 2 principal components for visualization.

6. Elbow Method

- K values from 1 to 10 are evaluated using inertia. - The elbow plot is saved to `charts/elbow\_method.png`.

7. Silhouette Analysis

- Silhouette scores for $K \geq 2$ are computed and saved to `charts/silhouette\_score.png`. - The pipeline chooses a candidate K based on the highest silhouette score.

8. K-Means Clustering

- Final KMeans model is trained and saved to `models/kmeans\_model.pkl`. - Segmented dataset is saved to `outputs/segmented\_customers.csv`.

9. Customer Personas

- Personas are generated by summarizing numeric feature averages per cluster and saved to `outputs/customer\_personas.csv`. - A visual table is saved to `charts/customer\_personas.png`.

10. Business Insights

- The report highlights the largest segment, the highest-spending segment, and recommended marketing strategies. These are printed and saved in the notebook output.

Personas Summary (computed)

- **Cluster 2 (Largest, n=546):** Low average `Quantity` (~1.68), moderate `UnitPrice` (~345.29), lower `ItemsInCart` (~3.77), average `TotalPrice` ≈ 569.68. This is the largest segment by count. - **Cluster 1 (High-spend, n=348):** Moderate `Quantity` (~4.04), highest `UnitPrice` (~534.65), moderate `ItemsInCart` (~6.78), highest `TotalPrice` ≈ 2135.89. This segment drives the highest revenue per order. - **Cluster 0 (Frequent buyers, n=306):** Moderate `Quantity` (~3.97), lower `UnitPrice` (~173.56), highest `ItemsInCart` (~7.07), `TotalPrice` ≈ 687.68. Likely frequent purchasers buying multiple lower-priced items.

Business Insights & Recommendations

- **Largest segment (Cluster 2):** Focus on retention campaigns — loyalty programs, personalized communications, and lifecycle email nudges. Consider targeted discounts to increase basket size without eroding margin. - **Highest spending customers (Cluster 1):** High-value customers — prioritize premium offers, early access to new products, and VIP service. Consider cross-sell and bundling to increase lifetime value. - **Frequent buyers (Cluster 0):** Candidates for subscription-based offers or bulk discounts. Encourage repeat purchases with reminders and time-limited promotions. - **Marketing Channels:** Review `ReferralSource` and `CouponCode` columns (present in `segmented\_customers.csv`) to attribute channels; prioritize channels that feed high-value segments. - **Next Steps:** A/B test targeted promotions per segment, run uplift modeling for marketing actions, and consider CLTV modeling to prioritize retention spend.

11. Conclusion

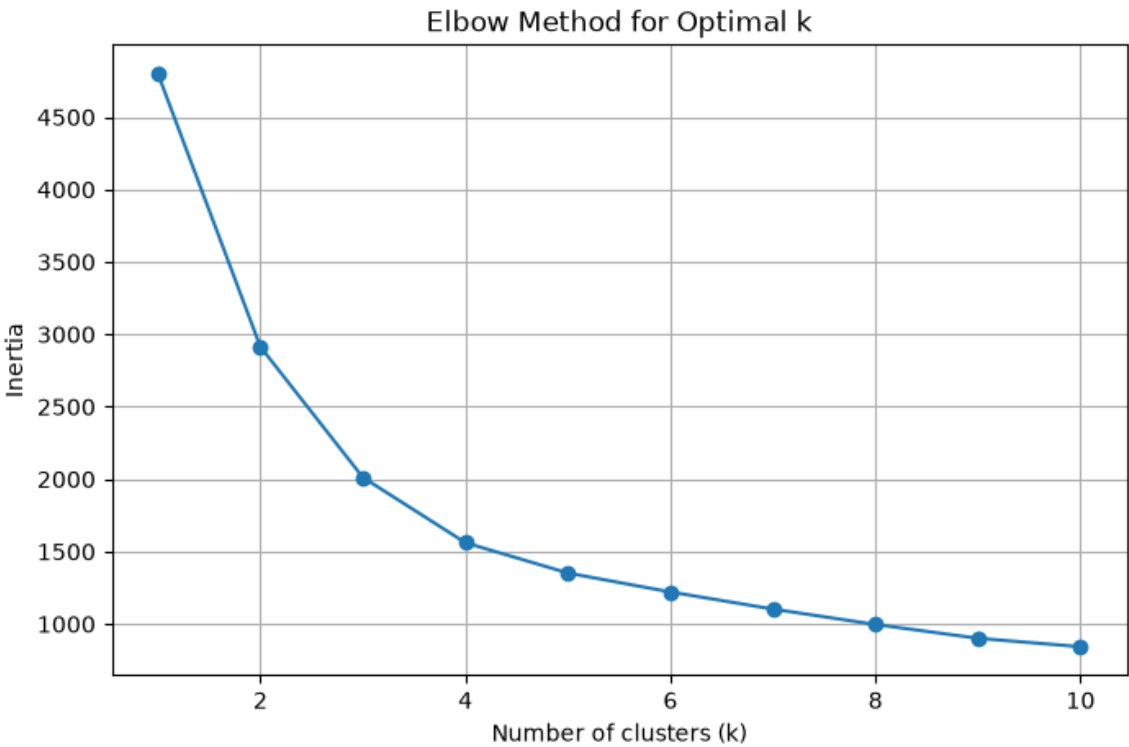
- The repository is ready for presentation as an internship portfolio artifact. For production deployments, consider more robust missing-value strategies, feature engineering, hyperparameter tuning, and running on larger datasets. Generated: 2026-06-15

KPI	Value
Dataset size	1200 rows × 14 cols
Features (sample)	Quantity, UnitPrice, ItemsInCart, TotalPrice
Missing values	309
Models trained	2

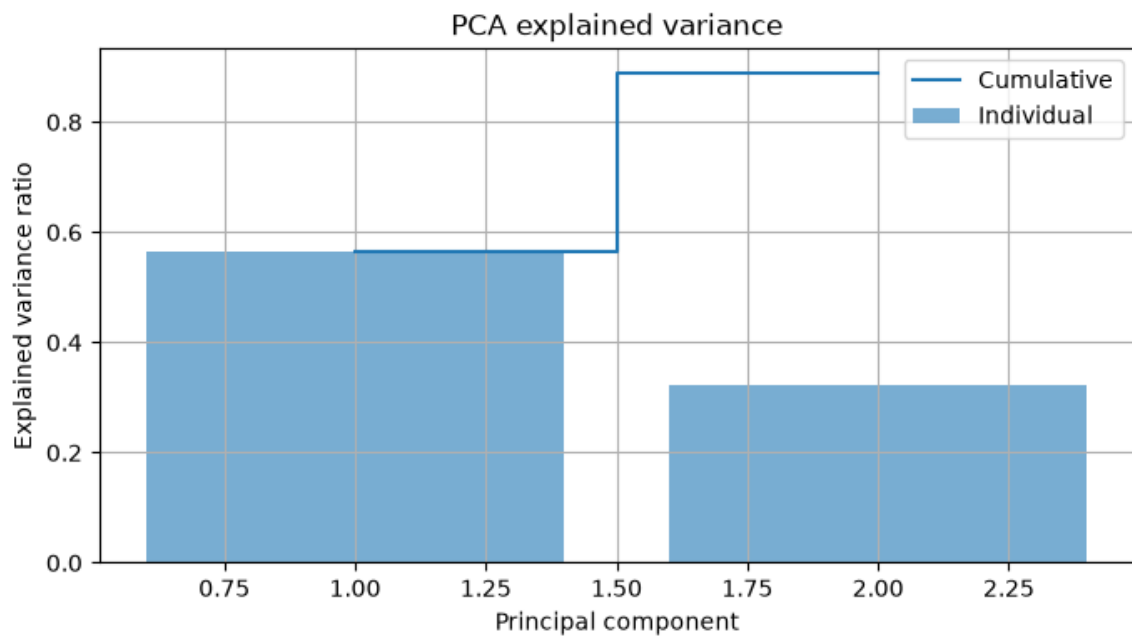
customer personas

Customer Personas					
cluster	Quantity	UnitPrice	ItemsInCart	TotalPrice	count
309673202614379078.5	5977124183000686274509803027.6	756535947712	306.0		
110373563218390534.6	517816091964758620689652135.8	926436781608	348.0		
126076556776556347.2	87893772893777472527472527969.6	750366300366	546.0		

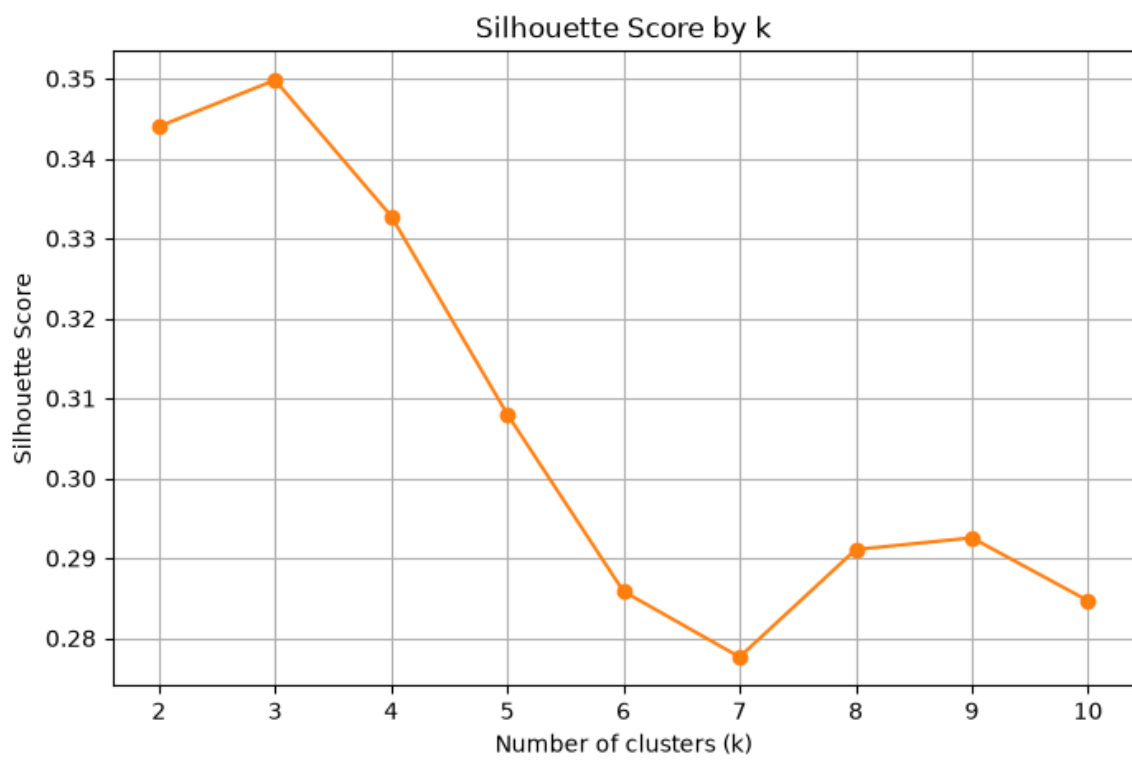
elbow method



pca variance explained



silhouette score



Appendix

Week 3 — Customer Segmentation Report

1. Executive Summary

This project implements a customer segmentation pipeline using PCA and K-Means clustering. It includes data cleaning, feature scaling, dimensionality reduction, cluster validation (elbow & silhouette), final clustering, and business-facing customer personas.

2. Dataset Overview

- The pipeline automatically loads a dataset from `Week\_3\_Customer\_Segmentation/data/`.
- Supported formats: CSV and Excel (.xlsx).
- The notebook prints dataset shape, columns, data types, missing value summary, and duplicates.

3. Data Cleaning

- Missing values are handled with simple imputation (dropping or filling where appropriate).
- Duplicate rows are removed.
- Cleaned dataset saved to `outputs/cleaned\_customers.csv`.

4. Feature Selection

- Numeric features are automatically selected for clustering.
- Identifiers (customer IDs) are removed from the feature matrix.

5. PCA Analysis

- PCA is applied to the scaled numeric features.
- The explained variance plot is saved to `charts/pca\_variance\_explained.png`.
- Data reduced to 2 principal components for visualization.

6. Elbow Method

- K values from 1 to 10 are evaluated using inertia.
- The elbow plot is saved to `charts/elbow\_method.png`.

7. Silhouette Analysis

- Silhouette scores for $K \geq 2$ are computed and saved to `charts/silhouette\_score.png`.
- The pipeline chooses a candidate K based on the highest silhouette score.

8. K-Means Clustering

- Final KMeans model is trained and saved to `models/kmeans\_model.pkl`.
- Segmented dataset is saved to `outputs/segmented\_customers.csv`.

9. Customer Personas

- Personas are generated by summarizing numeric feature averages per cluster and saved to `outputs/customer\_personas.csv`.
- A visual table is saved to `charts/customer\_personas.png`.

10. Business Insights

- The report highlights the largest segment, the highest-spending segment, and recommended marketing strategies. These are printed and saved in the notebook output.

Personas Summary (computed)

- **Cluster 2 (Largest, n=546):** Low average `Quantity` (~1.68), moderate `UnitPrice` (~345.29), lower `ItemsInCart` (~3.77), average `TotalPrice` ≈ 569.68. This is the largest segment by count.
- **Cluster 1 (High-spend, n=348):** Moderate `Quantity` (~4.04), highest `UnitPrice` (~534.65), moderate `ItemsInCart` (~6.78), highest `TotalPrice` ≈ 2135.89. This segment drives the highest revenue per order.